

Mutation Authority as a Predictive Risk Factor: Empirical Evidence from 83 Blockchain Deployments

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Abstract—We present the Blockchain Risk Evaluation Model (BREM), a multi-domain framework that scores blockchain implementations across nine canonical dimensions organized in a Structure-Process-Persistence matrix applied to three coordination layers: Network (machine-to-machine), System State (human-to-machine), and Law (human-to-human). Applied to 83 projects spanning enterprise consortia, central bank pilots, and DeFi protocols, the model identifies a failure threshold at an overall score of 2.5 or above, achieving 92.5% sensitivity ($F1 = 0.902$) and 100% specificity among resolved projects. The central diagnostic variable is mutation authority (sm), measuring the scope of discretion any single entity retains over protocol rules, data structures, and execution environment. Of 83 projects, 76 score sm at 3 or above, and no project with sm at 2 or below has failed. The model identifies correlated risk propagation chains converging on economic nonviability, and a DeFi-specific cascade reversal that produces the same failure mechanism through a different entry point. The three false negatives are architecturally sound systems that failed commercially, suggesting the model captures systemic coordination risk more reliably than market-driven failure.

Index Terms—blockchain risk, DeFi, mutation authority, failure prediction, risk framework, systemic risk

I. INTRODUCTION

Decentralized Finance (DeFi) has produced both transformative innovation and catastrophic failure. The collapse of Terra/Luna (\$40B in market capitalisation destroyed [13]), FTX (\$8B in customer funds unaccounted [14]), and Celsius (\$4.7B in frozen deposits [15]) demonstrated that DeFi risk assessment remains largely post-hoc—failures are diagnosed after capital is destroyed. Enterprise blockchain has experienced a parallel pattern: the ASX CHESS replacement consumed over \$250M before abandonment [6], while TradeLens and Marco Polo were discontinued despite significant institutional investment, for reasons that appear, in retrospect, structurally predictable.

The central challenge is that existing risk approaches evaluate isolated dimensions—smart contract auditing [1], economic mechanism design [2], or regulatory compliance—without integrating technical architecture, governance design, and legal enforceability into a unified assessment. We therefore propose an integrated framework that scores these three layers in one matrix.

This paper presents BREM (Blockchain Risk Evaluation Model), which evaluates blockchain systems across nine canonical risk dimensions using a Structure $\times$ Process $\times$

Persistence (SPP) matrix applied at three coordination layers. We demonstrate its predictive accuracy across an 83-project dataset spanning enterprise, institutional, and DeFi deployments, and identify *mutation authority*—the scope of discretion retained over protocol rules—as the single most diagnostic risk variable.

The rest of this paper is organized as follows. Section II reviews related work. Section III presents the BREM framework and scoring methodology. Section IV describes the dataset and validation approach. Section V reports results. Section VI discusses implications and limitations. Section VII concludes.

A. Contributions

- 1) A novel 3×3 risk evaluation framework (SPP $\times$ SPP) that integrates network architecture, system governance, and legal enforceability into nine canonical risk dimensions;
- 2) An 83-project scored dataset spanning enterprise consortia, central bank pilots, and DeFi protocols, with 92.5% sensitivity and 100% specificity among resolved projects at the ≥ 2.5 threshold;
- 3) Identification of mutation authority (sm) as the defining risk variable, with empirical evidence that no project with $sm \leq 2$ has failed in the observed sample;
- 4) Discovery of a DeFi-specific cascade reversal ($sf \rightarrow sm$) that produces the same failure mechanism through a different causal entry point.

II. RELATED WORK

A. DeFi Risk Frameworks

DeFi risk analysis has advanced significantly in recent years. Gudgeon et al. [3] developed a framework for analyzing DeFi protocol exploits through economic and technical lenses. Werner et al. [4] provided a systematic classification of DeFi risks across smart contract, economic, and governance dimensions. Xu et al. [5] proposed quantitative metrics for DeFi protocol resilience. However, these frameworks typically focus on protocol-specific risks rather than cross-domain structural patterns that predict failure across both DeFi and enterprise implementations.

B. Blockchain Failure Analysis

Post-mortem analyses of blockchain project failures have identified recurring patterns. The ASX CHESS replacement

failure has been documented as an architecture-scalability problem [6]. Catalini and Gans [7] examined the economics of blockchain adoption, identifying coordination costs as a key barrier. Carson et al. [8] surveyed enterprise blockchain adoption, noting high failure rates without providing a predictive framework. Wüst and Gervais [11] proposed a structured decision model for whether blockchain is appropriate, but did not quantify risk across deployed systems.

C. Governance and Centralisation Risk

The concept of protocol governance as a risk factor has been explored by Barbereau et al. [9], who examined governance token concentration in DeFi. Fritsch et al. [10] analyzed voting power distribution in DAOs. Walch [12] identified software governance as a source of centralization risk in ostensibly decentralized systems. Our contribution extends this line of work by formalizing *mutation authority*—the operational scope of discretion over protocol rules, not merely token-weighted voting power—as a scored dimension within a multi-domain framework, and demonstrating its predictive power across 83 projects spanning both enterprise and DeFi.

III. THE BREM FRAMEWORK

A. SPP $\times$ SPP Architecture

BREM applies a Structure $\times$ Process $\times$ Persistence (SPP) lens at two levels. At the macro level, the three coordination layers *are* SPP: Network (M2M) is the **structure** of the coordination system; System State (H2M) is the **process** operating on that structure; and Law (H2H) is the **persistence** question—whether the system survives contact with legal and institutional reality.

At the micro level, each layer is interrogated through the same SPP lens, producing nine canonical cells (Table I).

TABLE I
THE BREM 3 $\times$ 3 EVALUATION MATRIX

	Structure	Process	Persistence
Network (M2M)	na	nc	ns
System State (H2M)	se	sm	sf
Law (H2H)	ls	lr	lp

Each cell abbreviation denotes: na = Architecture, nc = Consensus, ns = Scalability, se = Execution, sm = Mutation Authority, sf = Economic Fitness, ls = Standing, lr = Remedy, lp = Liability Precision.

B. Scoring: The 0–4 Scale

Each cell is scored 0–4, where 0 represents negligible risk (protocol-constrained, no discretionary authority) and 4 represents critical risk (fully centralized control or fundamental architectural failure). The scale is calibrated to each domain: **Network** measures protocol determinism (0 = independent state, no shared mutable data; 4 = shared mutable global state with sequential execution bottlenecks). **System State** measures mutation authority (0 = immutable protocol, no admin keys; 4 = database-administrator-level discretion). **Law** measures

enforceability (0 = clear standing, established remedies, identifiable liable parties; 4 = non-attributable participants with no legal recourse).

Each cell is assessed through five sub-questions (45 total across all nine cells), producing granular scores that aggregate to cell-level and domain-level means. The full scoring rubric, with sub-question definitions and score-level criteria, is published with the dataset.

C. Mutation Authority (sm): The Diagnostic Variable

Mutation authority measures the scope of discretion any entity retains over: (1) protocol rules and consensus parameters; (2) data structures and state management; (3) execution environment and smart contract upgradeability; (4) network parameters and access control; and (5) economic rules and fee structures.

This variable captures whether a system has committed to protocol-constrained state transition (sm = 0) or retains human override capability (sm $\geq$ 3). Unlike governance token concentration [9] or voting power distribution [10], mutation authority measures *operational* discretion: whether admin keys exist, whether contracts are upgradeable, whether parameters can be altered unilaterally. It is the architectural boundary between a blockchain and a database with distributed witnesses.

IV. DATASET AND METHODOLOGY

A. Dataset Composition

The dataset comprises 83 blockchain implementations:

- **Enterprise/Institutional** (n=71): Projects from financial services (securities settlement, FX, payments), trade finance, supply chain, healthcare, energy, government digital identity, and central bank digital currencies, spanning 2015–2026.
- **DeFi** (n=12): Protocol collapses (Terra/Luna, Celsius, FTX, Iron Finance, Wonderland), survivors (Uniswap V2, Aave, MakerDAO/Sky, Compound), and distressed-but-operational protocols (Axie Infinity/Ronin, Solend, Olympus DAO).

Of the 83 projects, 62 have reached definitive outcomes (*resolved*), while 21 remain in pilot, research, or early development phases (*unresolved*). Each project is documented with structured characteristics (permission model, consensus family, architecture type, consortium size) and evidence-backed scoring justifications derived from public sources.

B. Scoring Process

Each project is scored across 45 sub-questions (5 per cell), each yielding a 0–4 integer score. Sub-question scores are averaged to produce cell scores, cell scores are averaged within rows to produce domain scores, and the overall BREM score is the mean of all nine cell scores.

Scores are derived from publicly available evidence: technical whitepapers, protocol documentation, governance forum records, on-chain contract analysis, regulatory filings, and

post-mortem reports. Each score is accompanied by an evidence citation identifying the source and the specific characteristic being assessed.

The scoring process proceeded in two phases. In Phase 1, 10 anchor projects spanning the full risk spectrum—confirmed failures (ASX CHESS, Terra/Luna, TradeLens), mid-range pilots (Project Acacia, BIS Mandala), and production survivors (Uniswap V2)—were scored independently across all 45 sub-questions with detailed evidence documentation. In Phase 2, the remaining 73 projects were scored using the same rubric, with anchor scores serving as calibration references to ensure consistent score-level application across projects with similar architectures and governance structures.

Hindsight bias mitigation. The scoring rubric assesses architectural and governance properties observable at the *design stage*, not outcomes. For example, the *sm* sub-questions ask whether admin keys exist, whether contracts are upgradeable, and whether governance can alter protocol parameters—all verifiable from protocol documentation before any failure occurs. To further test for outcome bias, 10 projects were scored blind to their current status by a second assessor; concordance was 89% within ± 0.5 points per cell.

C. Statistical Validation

Internal consistency was validated through Spearman rank correlation analysis between architecturally linked cells. The correlation between Architecture (*na*) and Scalability (*ns*) is $\rho = 0.606$ ($p < 0.001$), and between Scalability (*ns*) and Economic Fitness (*sf*) is $\rho = 0.566$ ($p < 0.001$), consistent with the causal chain predicted by the Architecture Death Spiral cascade (Section V-C). These moderate-to-strong correlations are expected: while the cascade model predicts directional influence, each cell also captures independent risk variation.

The status-to-score gradient provides external validation: not-failed systems average 2.12 ($\sigma = 0.45$) across all nine cells, while confirmed failures average 3.04 ($\sigma = 0.38$)—a 0.92-point separation (Cohen’s $d = 2.19$, very large effect). A Mann-Whitney U test confirms the distributions are significantly different ($p < 0.001$).

Inter-rater reliability, measured on a 10-project subsample scored by two independent assessors, yielded Cohen’s $\kappa = 0.81$ (substantial agreement) at the cell level, rising to $\kappa = 0.91$ at the domain level.

V. RESULTS

A. Failure Threshold

A natural failure boundary emerges at an overall BREM score of ≥ 2.5 (Table II).

Among the 62 resolved projects, specificity is 100%: every project above the threshold that has reached a definitive outcome has failed or entered sustained distress. The five “false positives” in the full dataset are all unresolved projects still in pilot or development; these represent predictions that will provide out-of-sample validation as projects mature.

The three false negatives—HSBC FX Everywhere (2.33), NASDAQ Linq (2.33), and SETL (2.33)—are projects with

TABLE II
THRESHOLD PERFORMANCE AT OVERALL SCORE ≥ 2.5

Metric	Full (n=83)	Resolved (n=62)
Sensitivity	92.5%	92.5%
Specificity	88.4%	100.0%
Precision	88.1%	100.0%
F1 Score	0.902	0.961

sound technical architecture that failed for commercial rather than systemic reasons. All three have low architecture risk ($na \leq 2$) and moderate scalability risk, which pulls the overall score below threshold even when governance ($sm \geq 3$) and economic fitness ($sf \geq 3$) are elevated. This pattern suggests the model captures systemic coordination failure more reliably than pure market-driven failure—a meaningful distinction because it identifies the type of failure most closely linked to architectural design choices.

B. Mutation Authority Distribution

Table III presents the distribution of mutation authority scores.

TABLE III
MUTATION AUTHORITY (SM) DISTRIBUTION (N=83)

sm	Count	%	Failed	Failure Rate
0	1	1.2%	0	0%
1	0	0%	—	—
2	6	7.2%	0	0%
3	42	50.6%	15	35.7%
4	34	41.0%	25	73.5%

The single $sm = 0$ project is Uniswap V2, a permissionless DeFi protocol with immutable core contracts, no admin keys, and no upgrade mechanism. Zero enterprise projects score below $sm = 2$. No project with $sm \leq 2$ has failed in the observed sample: the failure rate at $sm = 3$ is 35.7%, rising to 73.5% at $sm = 4$, while projects with bounded or absent mutation authority show a 0% failure rate ($n = 7$).

C. Failure Cascades

Cell-level analysis identifies three correlated risk propagation chains (Table IV).

TABLE IV
RISK PROPAGATION CHAINS AND SUPPORTING EVIDENCE

Cascade	Path	ρ	p
Architecture $\rightarrow$ Scaling	na $\rightarrow$ ns	0.606	<0.001
Scaling $\rightarrow$ Economics	ns $\rightarrow$ sf	0.566	<0.001
Mutation $\rightarrow$ Execution	sm $\rightarrow$ se	0.462	<0.001
Execution $\rightarrow$ Standing	se $\rightarrow$ ls	0.489	<0.001

The strongest chain runs na $\rightarrow$ ns $\rightarrow$ sf: architecture constrains scalability, which prevents economic viability. Economic fitness (*sf*) shows the largest effect size between failed and not-failed projects (Cohen’s $d = 2.77$), followed by execution

(se, $d = 1.72$) and scalability (ns, $d = 1.29$). This suggests that most failure pathways converge on economic nonviability, regardless of which domain initiates the cascade.

D. DeFi Cascade Reversal

DeFi failures exhibit the same propagation chains through reversed entry points (Table V).

TABLE V
ENTERPRISE VS. DeFi CASCADE COMPARISON

	Enterprise	DeFi
Primary cascade	na→ns→sf	sf→sm
Entry domain	Network	System State
Mechanism	Architecture constrains scalability	Economics expose governance
Law domain	Low risk (regulated)	High risk (no entity)
sm role	Cause	Effect

In enterprise deployments, mutation authority *causes* architectural constraints: retained governance discretion selects permissioned, centralized designs. In DeFi, economic failure *exposes* mutation authority: unsustainable tokenomics create pressure that governance discretion is then used to address, often destructively. Terra/Luna exemplifies this: the UST depeg (sf failure) exposed governance discretion that was exercised through minting billions of LUNA, accelerating the death spiral.

The DeFi sub-dataset produces separation with zero overlap: all 5 failures score ≥ 2.67 overall; all 7 non-failed projects score ≤ 2.33 . The 0.34-point gap between the highest non-failed DeFi project (Olympus DAO, 2.33) and the lowest failed (Terra/Luna, 2.67) straddles the 2.5 threshold.

E. Mutation Authority as Structural Divider

The most striking distributional finding is not domain-level averages but the sm variable itself. All 40 failures have $sm \geq 3$; all 7 projects with $sm \leq 2$ have survived ($\chi^2 = 5.16$, $p = 0.023$). Enterprise projects average $sm = 3.42$ compared to DeFi’s $sm = 2.58$, reflecting institutional requirements for discretionary override. The enterprise consensus mechanism score (nc = 2.89) is also notably higher than DeFi’s (nc = 1.83), consistent with the observation that enterprise deployments select centralized consensus to satisfy governance constraints.

F. Illustrative Cases

Terra/Luna (Overall: 2.67, Failed). The algorithmic stablecoin UST maintained its peg through an elastic supply mechanism. When the UST depeg began in May 2022, the sf failure (unsustainable economic model, sf = 4) exposed governance discretion (sm = 3): the Terra team minted billions of LUNA tokens attempting to restore the peg, accelerating the death spiral and destroying approximately \$40B in value.

Uniswap V2 (Overall: 1.22, Survivor). The sole sm = 0 project. Core contracts are immutable—deployed without admin keys, upgrade mechanisms, or pause functions. Governance exists only for the UNI token’s treasury allocation and

has no authority over the deployed protocol. This architectural commitment to immutability is associated with survival through multiple market crises.

ASX CHESS Replacement (Overall: 3.78, Failed). The Australian Securities Exchange invested over \$250M across seven years to replace its clearing system with a blockchain-based solution. The architecture selected shared mutable global state, triggering the na→ns→sf cascade: the system could not handle concurrent clearing volumes, and the project was abandoned in 2023.

VI. DISCUSSION

A. The Institutional Incompatibility Proposition

The empirical pattern supports a proposition: *institutional blockchain deployments converge toward mutable governance structures, and therefore away from blockchain’s defining property of protocol-constrained state transition.*

This convergence is driven by structural requirements: regulatory compliance mandates the ability to freeze assets, reverse transactions, and enforce sanctions; governance structures require discretionary override; operational processes assume mutable records. These requirements collectively guarantee $sm \geq 2$ for any institutional deployment. The result is a system that uses blockchain vocabulary and infrastructure while preserving every discretionary mechanism that blockchain was designed to remove—what we term a “database with distributed witnesses.”

Enterprise computing has, since EDI in the 1980s, operated under what we term the *simulation doctrine*: digital systems simulate events (purchase orders, settlement instructions) that humans later verify. A properly constructed blockchain transaction *is* the state change. When institutions adopt blockchain while preserving the simulation doctrine, they produce systems that are neither efficient databases nor genuine blockchains.

B. Implications for DeFi Resilience

The BREM framework is designed as a practical instrument. A risk analyst, protocol designer, or regulator can score a new or existing protocol across the nine cells using the published rubric, producing a risk profile in under an hour. This makes several DeFi risk management applications immediately viable:

Predictive risk scoring. New protocols can be evaluated against the 9-cell matrix before launch. The scoring rubric assesses architectural properties observable at design time—admin key presence, contract upgradeability, consensus mechanism, economic model sustainability—producing a composite score that locates the project relative to the 2.5 failure threshold. This provides a structured alternative to ad-hoc due diligence.

Systemic risk monitoring. The sm variable provides a single, measurable indicator of governance centralization risk. Protocols with $sm \geq 3$ are structurally vulnerable to the sf→sm cascade when economic conditions deteriorate. Continuous monitoring of sm across a portfolio of DeFi positions would

flag governance centralization before it manifests as a crisis—functioning as an early-warning system for the type of failure observed in Terra/Luna and Celsius.

Stress testing and simulation. The three propagation chains (na→ns→sf, sm→se→ls, sf→sm) provide a structural basis for DeFi stress tests. Rather than simulating price shocks alone, stress scenarios can model how architectural constraints propagate through the cascade—e.g., what happens to a protocol’s sf score when ns degrades under load, or how economic stress (sf) exposes governance discretion (sm).

Responsible governance design. The framework provides empirical evidence that protocol immutability is achievable (Uniswap V2, sm = 0) and that retained discretion is exploitable under economic stress (Terra/Luna, sm = 3). Protocol designers can use the matrix to identify which cells drive their risk profile above threshold and make targeted architectural changes before deployment.

C. Threats to Validity

Construct validity. The 0–4 scoring scale involves expert judgment. While calibrated to domain-specific criteria and validated through inter-rater reliability ($\kappa = 0.81$ cell-level, 0.91 domain-level), scores are not purely automated measurements.

Internal validity. Knowledge of project outcomes may influence scoring despite the design-stage rubric. The blind-scoring test (89% concordance, $n = 10$) mitigates but does not eliminate this concern.

External validity. The DeFi sub-dataset ($n = 12$) is small. DeFi-specific findings (cascade reversal, clean separation) should be treated as preliminary until validated on a larger sample. The three false negatives suggest the model is better calibrated for systemic coordination risk than for commercial viability.

Temporal dynamics. The model does not incorporate how risk profiles evolve as projects mature. A project scored at deployment may shift as governance structures change. Longitudinal tracking would strengthen the framework.

D. Future Work

Future work includes: (1) expanding the DeFi dataset to improve statistical power for cascade analysis; (2) developing AI-driven automated scoring tools that derive BREM cell scores from on-chain governance data, contract metadata, and protocol documentation using anomaly detection and NLP techniques; (3) longitudinal tracking of BREM scores as an early-warning system, where changes in sm or sf trigger alerts before cascade propagation; (4) integration with DeFi stress testing frameworks, using the propagation chains as structural models for scenario simulation; and (5) cross-chain risk analysis, extending the matrix to evaluate bridge protocols and multi-chain deployments where mutation authority may be distributed across governance domains. The 21 unresolved projects provide a natural out-of-sample validation cohort as they mature.

VII. CONCLUSION

BREM provides a structured, multi-domain framework for blockchain risk evaluation with demonstrated predictive accuracy across 83 projects. The central finding—that mutation authority is the defining variable separating protocol-constrained systems from governed databases—has implications for both DeFi protocol design and enterprise blockchain strategy.

The framework achieves 92.5% sensitivity at the ≥ 2.5 threshold with 100% specificity among resolved projects, identifies correlated risk propagation chains converging on economic nonviability, and reveals a DeFi-specific cascade reversal that produces the same structural failure through a different entry point. No project with $sm \leq 2$ has failed in the observed sample, suggesting that the architectural boundary between blockchain and database is observable, measurable, and almost universally compromised.

For DeFi resilience specifically, the model provides a predictive instrument: protocols with high mutation authority are structurally vulnerable to governance exploitation under economic stress. The framework’s principal limitation—that it captures systemic coordination risk better than commercial failure—is also its principal strength: it identifies the failure mode most directly attributable to architectural design choices.

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DATA AVAILABILITY

The scored dataset (83 projects $\times$ 40 columns), scoring rubric, codebook, and per-project evidence summaries are available at: [https://github.com/\[redacted-for-review\]](https://github.com/[redacted-for-review]). We encourage independent replication and extension.

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